

The China Syndrome: Local Labor Market Effects of Import Competition in the US

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American Economic Review, October 2013

Outline

Introduction

Theoretical predictions

Measuring exposure to import competition

Results

- Adverse impact on manufacturing employment

- Limited worker reallocation between CZs

- Adjustments within trade-exposed CZs

Robustness checks

Stylized facts

- Pre-1990s limited impact trade on US labor
- Trend decline in US manufacturing employment
- Increase in import competition from China
- Varies across regions in the US
- Limited offsetting increase in demand for US exports

Import penetration and manufacturing employment

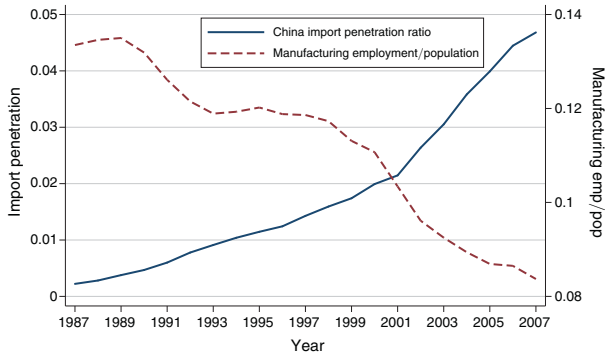
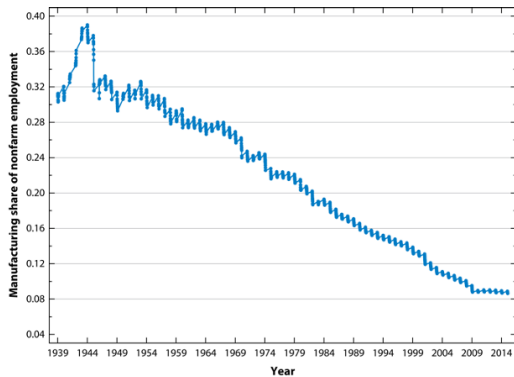


FIGURE 1. IMPORT PENETRATION RATIO FOR US IMPORTS FROM CHINA (*left scale*), AND SHARE OF US WORKING-AGE POPULATION EMPLOYED IN MANUFACTURING (*right scale*)

Long-run decline in US manufacturing employment



Autor DH, et al. 2016.
Annu. Rev. Econ. 8:205–40

Research question

- Impact of Chinese imports on US local labor markets (CZ)
 - CZs differently exposed because of industry specialization
- Analyze impact on non-wage variables
 - Employment, unemployment, participation, mobility, transfers
- Develop Shift-Share Instrumental Variable (SSIV)
 - Instrument US import competition ($\text{share} \times \text{shock}$) with imports (shock) by other countries

Key findings for trade-exposed CZs

- Lower wages and employment in (non-)manufacturing
- Higher unemployment and lower labor force participation
- Increased reliance on welfare benefits
- Limited evidence of emigration
- Significant distributional consequences of trade in reality!

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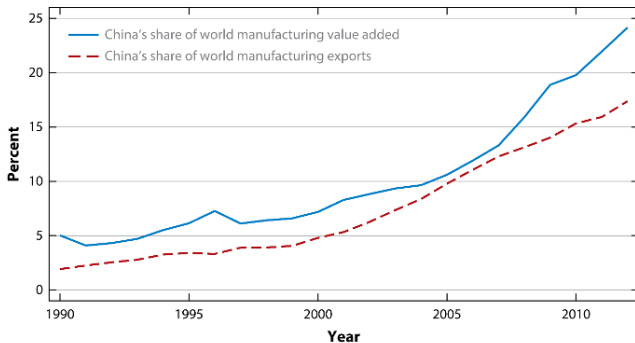
- Adjustments within trade-exposed CZs

Robustness checks

Exogenous shocks

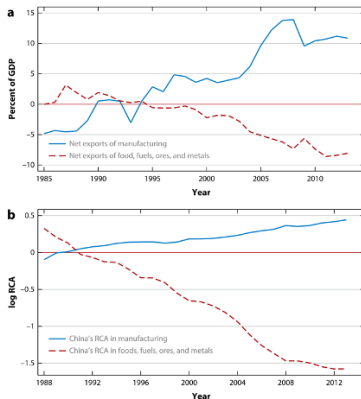
- China experienced:
 - Productivity growth (central planning → market economy)
 - Reduction in trade costs (accession to WTO in 2000)
- Opposing shocks in demand for goods in US CZs:
 - Increased competition in markets in which CZs sell output
 - Increased demand for goods in China

China's rise in world manufacturing



AR Autor DH, et al. 2016.
Annu. Rev. Econ. 8:205–40

China's revealed comparative advantage in manufacturing



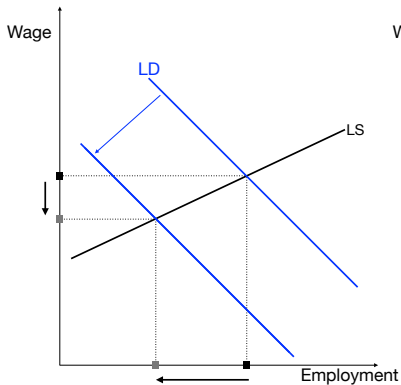
 Autor DH, et al. 2016.
Annu. Rev. Econ. 8:205–40

Impacts on labor demand and supply in US CZs

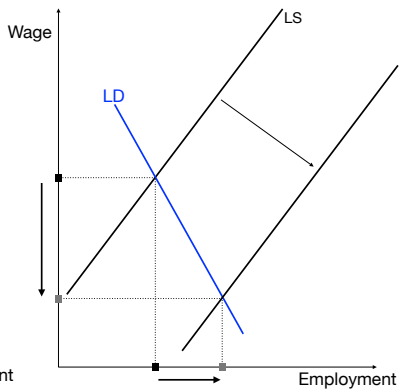
- Positive shock to China's export supply:
 - Decrease labor demand in traded goods
 - Increase labor supply and demand in non-traded sectors
- Positive shock to China's import demand:
 - Increase labor demand in traded goods
 - Decrease labor supply and increase labor demand in non-traded sectors
- Overall adverse effects driven by US trade deficit and regional specialization within US

Impact on manufacturers of imports

A. Manufacturing of imports

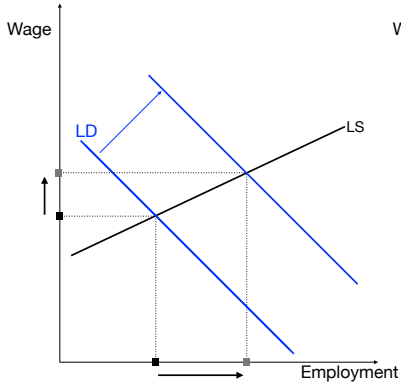


B. Non-manufacturing

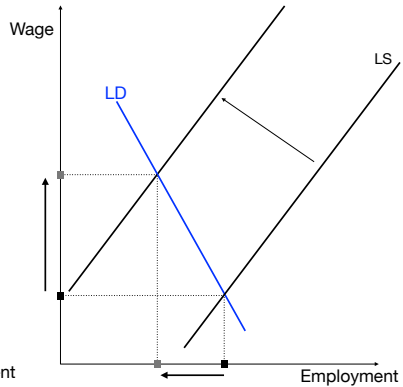


Impact on manufacturers of exports

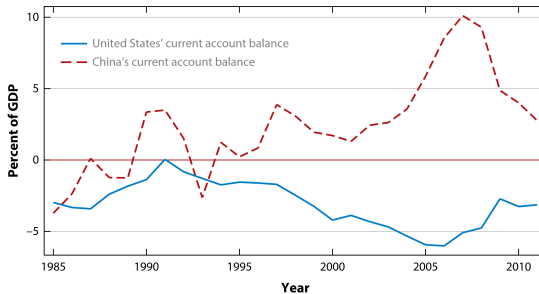
A. Manufacturing of exports



B. Non-manufacturing



Current account balances



Autor DH, et al. 2016.
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Measuring exposure to import competition

- Change in Chinese import exposure per worker in a CZ:

$$\Delta IPW_{zt} = \frac{\sum_i L_{izt} \Delta m_{it}}{L_{zt}} \text{ with } \Delta m_{it} = \frac{\Delta M_{it}}{L_{it}}$$

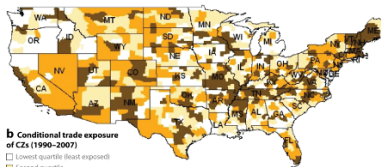
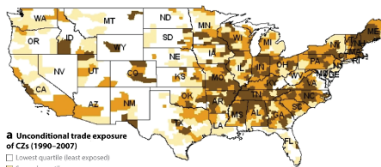
for CZ z and industry i

- Rewrite this as:

$$\Delta IPW_{zt} = \sum_i \frac{L_{izt}}{L_{zt}} \frac{\Delta M_{it}}{L_{it}} \quad (3)$$

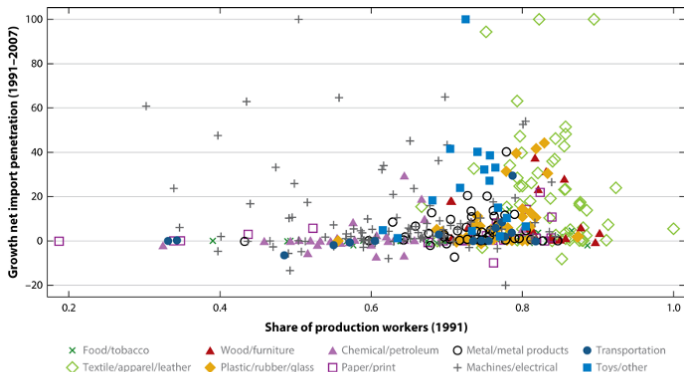
with L_{izt}/L_{zt} the degree of industrial specialization in z and $\Delta M_{it}/L_{it}$ the import value per worker in industry i

Trade exposure across CZs



 Autor DH, et al. 2016.
Annu. Rev. Econ. 8:205-40

Change in net import penetration by sector



Autor DH, et al. 2016.

Annu. Rev. Econ. 8:205-40

Shift-share instrument for import competition

- Δm_{it} could be correlated with unobserved US demand shocks
- Define the instrument:

$$\Delta IPW_{zt} = \sum_i \frac{L_{izt-1}}{L_{zt-1}} \frac{\Delta \tilde{M}_{it}}{L_{it-1}} \quad (4)$$

with $\Delta \tilde{M}_{it}$ Chinese imports into 8 other countries and $t - 1$ to account for expected employment impact of China shock at t

- Instrument predicts supply-driven changes in Chinese imports (e.g. China's SEZ, accession to WTO in 2000)

Empirical research design

- SSIV using a 2SLS estimator (using an instrument for the shift Δm_{it} and not for the share L_{izt}/L_{zt})
- Second stage:

$$\Delta Y_{zt}^m = \gamma_t + \beta_1 \Delta IP\hat{W}_{zt} + X'_{it}\beta_2 + \epsilon_{zt} \quad (5)$$

- First stage:

$$\Delta IPW_{zt} = \beta_3 \Delta IP\tilde{W}_{zt} + \tilde{\epsilon}_{zt}$$

- Standard errors are clustered at the state level to allow for spatial correlations across CZs

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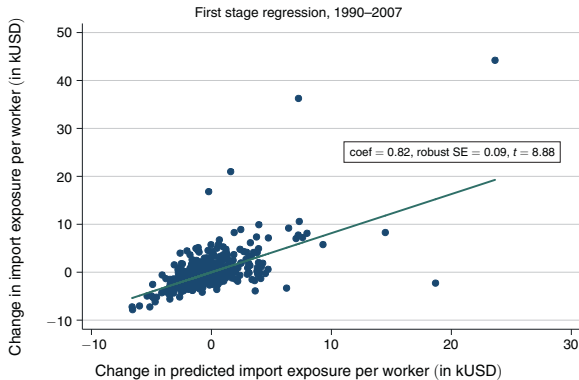
- Limited worker reallocation between CZs

- Adjustments within trade-exposed CZs

Robustness checks

First-stage estimates

Panel A. 2SLS first stage regression, full sample



Reduced-form estimates

Panel B. OLS reduced form regression, full sample

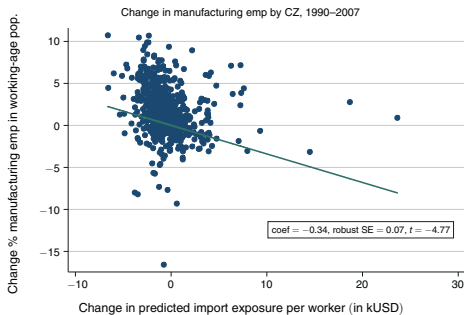


FIGURE 2. CHANGE IN IMPORT EXPOSURE PER WORKER AND DECLINE OF MANUFACTURING EMPLOYMENT:
ADDED VARIABLE PLOTS OF FIRST STAGE AND REDUCED FORM ESTIMATES

Notes: $N = 722$. The added variable plots control for the start of period share of employment in manufacturing industries. Regression models are weighted by start of period CZ share of national population.

Decline in manufacturing employment (no controls)

TABLE 2—IMPORTS FROM CHINA AND CHANGE OF MANUFACTURING EMPLOYMENT
IN CZs, 1970–2007: 2SLS ESTIMATES

Dependent variable: $10 \times$ annual change in manufacturing emp/working-age pop (in % pts)

	I. 1990–2007			II. 1970–1990 (pre-exposure)		
	1990–2000 (1)	2000–2007 (2)	1990–2007 (3)	1970–1980 (4)	1980–1990 (5)	1970–1990 (6)
(Δ current period imports from China to US)/worker	−0.89*** (0.18)	−0.72*** (0.06)	−0.75*** (0.07)			
(Δ future period imports from China to US)/worker				0.43*** (0.15)	−0.13 (0.13)	0.15 (0.09)

Notes: $N = 722$, except $N = 1,444$ in stacked first difference models of columns 3 and 6. The variable “future period imports” is defined as the average of the growth of a CZ’s import exposure during the periods 1990–2000 and 2000–2007. All regressions include a constant and the models in columns 3 and 6 include a time dummy. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

Decline in manufacturing employment (with controls)

TABLE 3—IMPORTS FROM CHINA AND CHANGE OF MANUFACTURING EMPLOYMENT
IN CZs, 1990–2007: 2SLS ESTIMATES
Dependent variable: 10 × annual change in manufacturing emp/working-age pop (in % pts)

	I. 1990–2007 stacked first differences					
	(1)	(2)	(3)	(4)	(5)	(6)
(Δ imports from China to US)/ worker	−0.746*** (0.068)	−0.610*** (0.094)	−0.538*** (0.091)	−0.508*** (0.081)	−0.562*** (0.096)	−0.596*** (0.099)
Percentage of employment in manufacturing _{−1}		−0.035 (0.022)	−0.052*** (0.020)	−0.061*** (0.017)	−0.056*** (0.016)	−0.040*** (0.013)
Percentage of college-educated population _{−1}				−0.008 (0.016)		0.013 (0.012)
Percentage of foreign-born population _{−1}				−0.007 (0.008)		0.030*** (0.011)
Percentage of employment among women _{−1}				−0.054** (0.025)		−0.006 (0.024)
Percentage of employment in routine occupations _{−1}					−0.230*** (0.063)	−0.245*** (0.064)
Average offshorability index of occupations _{−1}					0.244 (0.252)	−0.059 (0.237)
Census division dummies	No	No	Yes	Yes	Yes	Yes
	II. 2SLS first stage estimates					
	(1)	(2)	(3)	(4)	(5)	(6)
(Δ imports from China to OTH)/ worker	0.792*** (0.079)	0.664*** (0.086)	0.652*** (0.090)	0.635*** (0.090)	0.638*** (0.087)	0.631*** (0.087)
R ²	0.54	0.57	0.58	0.58	0.58	0.58

Notes: $N = 1,444$ (722 commuting zones $\times$ 2 time periods). All regressions include a constant and a dummy for the 2000–2007 period. First stage estimates in panel II also include the control variables that are indicated in the corresponding columns of panel I. Routine occupations are defined such that they account for 1/3 of US employment in 1980. The offshorability index variable is standardized to mean of 0 and standard deviation of 10 in 1980. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

Limited worker reallocation between CZs

TABLE 4—IMPORTS FROM CHINA AND CHANGE OF WORKING-AGE POPULATION
IN CZ, 1990–2007: 2SLS ESTIMATES
Dependent variables: Ten-year equivalent changes in log population counts (in log pts)

	I. By education level			II. By age group		
	All (1)	College (2)	Noncollege (3)	Age 16–34 (4)	Age 35–49 (5)	Age 50–64 (6)
<i>Panel A. No census division dummies or other controls</i>						
(Δ imports from China to US)/worker	−1.031** (0.503)	−0.360 (0.660)	−1.097** (0.488)	−1.299 (0.826)	−0.615 (0.572)	−1.127*** (0.422)
R^2	—	0.03	0.00	0.17	0.59	0.22
<i>Panel B. Controlling for census division dummies</i>						
(Δ imports from China to US)/worker	−0.355 (0.513)	0.147 (0.619)	−0.240 (0.519)	−0.408 (0.953)	−0.045 (0.474)	−0.549 (0.450)
R^2	0.36	0.29	0.45	0.42	0.68	0.46
<i>Panel C. Full controls</i>						
(Δ imports from China to US)/worker	−0.050 (0.746)	−0.026 (0.685)	−0.047 (0.823)	−0.138 (1.190)	0.367 (0.560)	−0.138 (0.651)
R^2	0.42	0.35	0.52	0.44	0.75	0.60

Notes: $N = 1,444$ (722 CZs $\times$ two time periods). All regressions include a constant and a dummy for the 2000–2007 period. Models in panel B and C also include census division dummies while panel C adds the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period commuting zone share of national population.

Limited reallocation of workers between CZ

- No evidence that shocks to local manufacturing lead to change in population
- Shocks to manufacturing too small to affect outcomes in broader CZ
- Shocks are substantial but no reallocation of workers because moving costs too high
- CZ absorbs shock through other employment in CZ and/or lower wages

Job loss in trade-exposed CZs

TABLE 5—IMPORTS FROM CHINA AND EMPLOYMENT STATUS OF WORKING-AGE POPULATION
WITHIN CZs, 1990–2007: 2SLS ESTIMATES
*Dependent variables: Ten-year equivalent changes in log population counts
and population shares by employment status*

	Mfg emp (1)	Non-mfg emp (2)	Unemp (3)	NILF (4)	SSDI receipt (5)
<i>Panel A. 100 × log change in population counts</i>					
(Δ imports from China to US)/worker	−4.231*** (1.047)	−0.274 (0.651)	4.921*** (1.128)	2.058* (1.080)	1.466*** (0.557)
<i>Panel B. Change in population shares</i>					
<i>All education levels</i>					
(Δ imports from China to US)/worker	−0.596*** (0.099)	−0.178 (0.137)	0.221*** (0.058)	0.553*** (0.150)	0.076*** (0.028)
<i>College education</i>					
(Δ imports from China to US)/worker	−0.592*** (0.125)	0.168 (0.122)	0.119*** (0.039)	0.304*** (0.113)	—
<i>No college education</i>					
(Δ imports from China to US)/worker	−0.581*** (0.095)	−0.531*** (0.203)	0.282*** (0.085)	0.831*** (0.211)	—

Notes: $N = 1,444$ (722 CZs × two time periods). All statistics are based on working age individuals (age 16 to 64). The effect of import exposure on the overall employment/population ratio can be computed as the sum of the coefficients for manufacturing and nonmanufacturing employment; this effect is highly statistically significant ($p \leq 0.01$) in the full sample and in all reported subsamples. All regressions include the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

Negative impact on average wage in trade-exposed CZs

TABLE 6—IMPORTS FROM CHINA AND WAGE CHANGES

WITHIN CZs, 1990–2007: 2SLS ESTIMATES

Dependent variable: Ten-year equivalent change in average log weekly wage (in log pts)

	All workers (1)	Males (2)	Females (3)
<i>Panel A. All education levels</i>			
(Δ imports from China to US)/worker	−0.759*** (0.253)	−0.892*** (0.294)	−0.614*** (0.237)
R^2	0.56	0.44	0.69
<i>Panel B. College education</i>			
(Δ imports from China to US)/worker	−0.757** (0.308)	−0.991*** (0.374)	−0.525* (0.279)
R^2	0.52	0.39	0.63
<i>Panel C. No college education</i>			
(Δ imports from China to US)/worker	−0.814*** (0.236)	−0.703*** (0.250)	−1.116*** (0.278)
R^2	0.52	0.45	0.59

Notes: $N = 1,444$ (722 CZs $\times$ two time periods). All regressions include the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

Negative impact on non-manuf. wage in trade-exposed CZs

TABLE 7—COMPARING EMPLOYMENT AND WAGE CHANGES IN MANUFACTURING
AND OUTSIDE MANUFACTURING, 1990–2007: 2SLS ESTIMATES

Dependent variables: Ten-year equivalent changes in log workers and average log weekly wages

	I. Manufacturing sector			II. Nonmanufacturing		
	All workers (1)	College (2)	Noncollege (3)	All workers (4)	College (5)	Noncollege (6)
<i>Panel A. Log change in number of workers</i>						
(Δ imports from China to US)/worker	−4.231*** (1.047)	−3.992*** (1.181)	−4.493*** (1.243)	−0.274 (0.651)	0.291 (0.590)	−1.037 (0.764)
R^2	0.31	0.30	0.34	0.35	0.29	0.53
<i>Panel B. Change in average log wage</i>						
(Δ imports from China to US)/worker	0.150 (0.482)	0.458 (0.340)	−0.101 (0.369)	−0.761*** (0.260)	−0.743** (0.297)	−0.822*** (0.246)
R^2	0.22	0.21	0.33	0.60	0.54	0.51

Notes: $N = 1,444$ (722 CZs $\times$ two time periods). All regressions include the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

Increased welfare payments in trade-exposed CZs

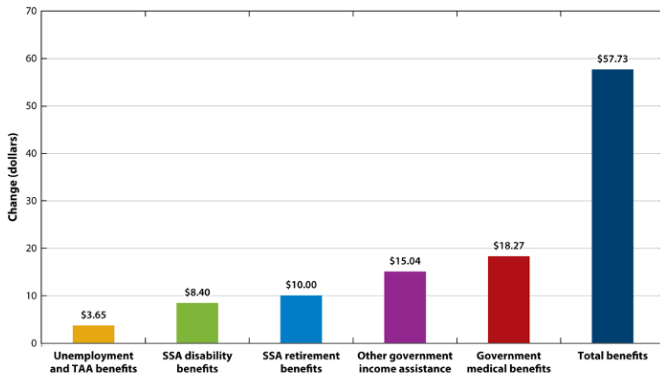
TABLE 8—IMPORTS FROM CHINA AND CHANGE OF GOVERNMENT TRANSFER RECEIPTS
IN CZs, 1990–2007: 2SLS ESTIMATES

Dep vars: Ten-year equivalent log and dollar change of annual transfer receipts per capita (in log pts and US\$)

	Total individual transfers (1)	TAA benefits (2)	Unem- ployment benefits (3)	SSA retirement benefits (4)	SSA disability benefits (5)	Medical benefits (6)	Federal income assist (7)	Educ/ training assist (8)
<i>Panel A. Log change of transfer receipts per capita</i>								
(Δ imports from China to US)/worker	1.01*** (0.33)	14.41* (7.59)	3.46* (1.87)	0.72* (0.38)	1.96*** (0.69)	0.54 (0.49)	3.04*** (0.96)	2.78** (1.32)
R^2	0.57	0.28	0.48	0.36	0.32	0.27	0.54	0.33
<i>Panel B. Dollar change of transfer receipts per capita</i>								
(Δ imports from China to US)/worker	57.73*** (18.41)	0.23 (0.17)	3.42 (2.26)	10.00* (5.45)	8.40*** (2.21)	18.27 (11.84)	7.20*** (2.35)	3.71*** (1.44)
R^2	0.75	0.28	0.41	0.47	0.63	0.66	0.53	0.37

Notes: $N = 1,444$ (722 CZs $\times$ two time periods), except $N = 1,436$ in column 2, panel A. Results for TAA benefits in column 2 are based on state-level data that is allocated to CZs in proportion to unemployment benefits. Unemployment benefits in column 3 include state benefits and federal unemployment benefits for civilian federal employees, railroad employees, and veterans. Medical benefits in column 6 consist mainly of Medicare and Medicaid. Federal income assistance in column 7 comprises the SSI, AFDC/TANF, and SNAP programs while education and training assistance in column 8 includes such benefits as interest payments on guaranteed student loans, Pell grants, and Job Corps benefits. The transfer categories displayed in columns 2 to 8 account for over 85 percent of total individual transfer receipts. All regressions include the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

Impact of \$1000 increase in exposure on welfare payments



Autor DH, et al. 2016.

Annu. Rev. Econ. 8:205–40

Decrease in HH income in trade-exposed CZs

TABLE 9—IMPORTS FROM CHINA AND CHANGE IN HOUSEHOLD INCOME, 1990–2007: 2SLS ESTIMATES

Dependent variable: Ten-year equivalent percentage and real dollar change in average and median annual household income per working-age adult (in %pts and US\$)

	Average HH income/adult by source				Median HH income/adult	
	Total (1)	Wage- salary (2)	Business invest (3)	SocSec + AFDC (4)	Total (5)	Wage- salary (6)
<i>Panel A. Percent change</i>						
(Δ imports from China to US)/worker	−1.48*** (0.36)	−2.14*** (0.59)	−0.51 (0.74)	2.12*** (0.58)	−1.73*** (0.38)	−2.32*** (0.51)
R^2	0.69	0.43	0.76	0.52	0.53	0.52
<i>Panel B. Dollar change</i>						
(Δ imports from China to US)/worker	−492.6*** (160.4)	−549.3*** (169.4)	40.1 (116.7)	17.3*** (4.3)	−439.9*** (112.7)	−476.5*** (122.2)
R^2	0.63	0.40	0.72	0.51	0.49	0.48

Notes: $N = 1,444$ (722 CZs $\times$ 2 time periods). Per capita household income is defined as the sum of individual incomes of all working-age household members (age 16–64), divided by the number of household members of that age group. Total income comprises wage and salary income; self-employment, business, and investment income; social security and welfare income; and income from other nonspecified sources. Social security and welfare income in column 4 includes social security retirement, disability, and supplementary income, aid to families with dependent children (AFDC), and general assistance. All regressions include the full vector of control variables from column 6 of Table 3. Robust standard errors in parentheses are clustered on state. Models are weighted by start of period CZ share of national population.

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Different measures of import competition

- Include competition in foreign markets
- Include imports of intermediate inputs
- Use net import from China (i.e. include export effects)
- Use gravity residual of relative productivity changes
- Use H-O factor content of US net imports from China

Conclusions

- Impact of Chinese import competition through reducing manufacturing employment
- Chinese import competition explains 21 percent of decline manufacturing employment in U.S. from 1991 to 2007
- Limited labor market adjustments across CZ
- In trade-exposed CZ, there is a decline in non-manufacturing wages, increase in unemployment and transfer payments

Further readings

- Autor, Dorn, Hanson, and Song. 2014. Trade adjustment: Worker Level Evidence. *Quarterly Journal of Economics*. 129(4):1799–860
- Autor, Dorn, and Hanson. 2016. The China Shock: Learning from Labor-Market Adjustment to Large Changes in Trade. *Annual Review of Economics*. 8:205–40
- Autor, Dorn, and Hanson. 2019. When Work Disappears: Manufacturing Decline and the Falling Marriage Market Value of Young Men. *American Economic Review: Insights*. 1(2):161-178
- Autor, Dorn, Hanson, and Majlesi. 2020. Importing Political Polarization? The Electoral Consequences of Rising Trade Exposure. *American Economic Review*. 110(10):3139–3183
- Autor, Dorn, and Hanson. 2021. On the Persistence of the China Shock. forthcoming in *Brookings Papers on Economic Activity*.
- Autor, Beck, Dorn, and Hanson. 2024. Help for the Heartland? The Employment and Electoral Effects of the Trump Tariffs.